

Stage 3B Physics Note: Modified-Cartoon Source Terms for the 5D Black String

GL-with-AI Project

June 15, 2026

1 Scope and Status

This note is a polished companion source for the Stage 3B implementation note. It is a derivation and validation roadmap only. No modified-cartoon source terms are implemented in `BlackStringToy`; the current scaffold does not evolve a five-dimensional black string.

The target reduction has gridded directions x, z and two hidden transverse directions w_1, w_2 , with `GR_SPACEDIM` = 4 and `CH_SPACEDIM` = 2. The primary risk is silent failure: a code may compile and run while omitting hidden-direction algebraic terms.

The Stage 1 source inspection found that the required 5D/SO(3) modified-cartoon CCZ4 source terms are absent from public GRChombo. Public CCZ4 contains dimension-parameterized algebra, but not the project-specific hidden-sphere source-term system required for `CH_SPACEDIM` = 2, `GR_SPACEDIM` = 4. This is a statement about the black-string target in this project, not a defect in public GRChombo's intended 3+1D use.

Thus Stage 3B is not merely checking whether these terms exist. It is planning the derivation, implementation, and validation of source terms that must be added in the project-specific layer.

2 Reduced Geometry

The transverse flat spatial metric can be written in spherical form as

$$dx^2 + x^2 d\Omega_2^2. \tag{1}$$

In the hidden Cartesian-like cartoon representation, flat transverse space has

$$\gamma_{ww} = 1. \tag{2}$$

The angular factor x^2 is supplied by coordinate reconstruction, not by setting h_{ww} or γ_{ww} to x^2 . The physical transverse sphere radius is

$$R = x\sqrt{\gamma_{ww}} = x\sqrt{\chi^{-1}h_{ww}}. \tag{3}$$

3 Spherical-to-Cartoon Map

Spherical object	Hidden/cartoon representation
$g_{\theta\theta} = x^2$	$\gamma_{ww} = 1$ plus external x^2 radius factor
$K_{\theta\theta} = x\beta^x$	$K_{ww} = \beta^x/x$
$\Gamma_{\theta\theta}^x = -x$	hidden algebraic source terms after contraction
$\Gamma_{x\theta}^\theta = 1/x$	explicit $1/x$ reduced source terms
angular trace contribution	two hidden contributions, one for each suppressed direction
diagonal trace	$T^x_x + T^z_z + 2T^w_w$

For this project the number of suppressed spatial directions is

$$N_{\text{hidden}} = \text{GR_SPACEDIM} - \text{CH_SPACEDIM} = 4 - 2 = 2, \quad (4)$$

so diagonal traces contain $2T^w_w$. Writing the factor this way helps avoid hard-coding the wrong multiplicity in future variants. The two hidden directions are equivalent here because of the imposed SO(3) spherical symmetry. In source-code notation, this is `N_hidden = GR_SPACEDIM - CH_SPACEDIM = 4 - 2 =`

4 GP Extrinsic-Curvature Check

For the natural GP shift

$$\beta^x = s\sqrt{\frac{r_0}{x}}, \quad (5)$$

and flat spatial metric

$$dl^2 = dx^2 + x^2 d\Omega_2^2 + dz^2, \quad (6)$$

the relevant spherical Christoffels are

$$\Gamma_{\theta\theta}^x = -x, \quad \Gamma_{\phi\phi}^x = -x \sin^2 \theta, \quad (7)$$

$$\Gamma_{x\theta}^\theta = \frac{1}{x}, \quad \Gamma_{x\phi}^\phi = \frac{1}{x}. \quad (8)$$

Using

$$K_{ij} = \frac{1}{2\alpha} (D_i \beta_j + D_j \beta_i - \partial_t \gamma_{ij}), \quad (9)$$

with $\alpha = 1$, one obtains

$$K_{xx} = \partial_x \beta_x = -\frac{s}{2} \sqrt{\frac{r_0}{x^3}}, \quad (10)$$

$$K_{\theta\theta} = x\beta^x = s\sqrt{r_0 x}, \quad (11)$$

$$K_{\phi\phi} = x\beta^x \sin^2 \theta, \quad (12)$$

$$K_{zz} = 0. \quad (13)$$

The $\sin^2 \theta$ factor cancels in the contracted contribution:

$$\gamma^{\phi\phi} K_{\phi\phi} = \frac{1}{x^2 \sin^2 \theta} (x\beta^x \sin^2 \theta) = \frac{\beta^x}{x}, \quad (14)$$

$$\gamma^{\theta\theta} K_{\theta\theta} = \frac{1}{x^2} (x\beta^x) = \frac{\beta^x}{x}. \quad (15)$$

This demonstrates why the two suppressed directions contribute equally after contraction and why the cartoon representation uses two copies of K^w_w . Mapping to hidden cartoon components gives

$$K_{ww} = \frac{\beta^x}{x} = s \sqrt{\frac{r_0}{x^3}}, \quad (16)$$

for each hidden direction, and

$$K = K^x_x + K^z_z + K^{w_1}_{w_1} + K^{w_2}_{w_2} = \frac{3s}{2} \sqrt{\frac{r_0}{x^3}}. \quad (17)$$

This check assumes the unperturbed $\chi = 1$ GP slice and remains subject to GRChombo’s possible overall K_{ij} sign convention.

5 Source-Term Categories

The eventual CCZ4 implementation must handle hidden metric variables h_{ww} and γ_{ww} , hidden extrinsic curvature A_{ww} and K_{ww} , full-dimensional traces with `GR_SPACEDIM` = 4, algebraic $1/x$ and $1/x^2$ replacements for suppressed derivatives, hidden Ricci tensor/scalar terms, conformal connection functions and Gamma constraints, gauge RHS terms, and regularization near small x .

In this Stage 3B target, hidden-direction multiplicities should be written as $N_{\text{hidden}} = \text{GR_SPACEDIM} - \text{CH_SPACEDIM} = 2$ equivalent $\text{SO}(3)$ contributions, not inferred from the number of grid axes alone.

Detailed CCZ4 RHS formulas are future work and require symbolic verification before implementation.

6 Validation Strategy

Validation should proceed in layers:

1. Symbolically derive reduced Christoffels, Ricci terms, traces, and CCZ4 RHS contributions, preferably with xAct/Mathematica or SymPy.
2. Unit-test Minkowski cartoon variables, the uniform GP black string, and the perturbed GL seed.
3. Track runtime diagnostics: Hamiltonian and momentum constraints, Θ , tracelessness, extrema of h_{ww} and γ_{ww} , source-term norms, regularity near small x , and horizon radius/area.
4. Run at least three resolutions and compare to exact/unperturbed solutions, because wrong source terms can converge to the wrong equations.

7 Failure Modes

Common silent failures include using `CH_SPACEDIM` = 2 in physical traces, missing the factor $N_{\text{hidden}} = \text{GR_SPACEDIM} - \text{CH_SPACEDIM} = 4 - 2 = 2$ for the two equivalent hidden directions, setting $h_{ww} = x^2$, confusing K_{ww} with $K_{\theta\theta}$, dropping $1/x$ or $1/x^2$ terms, failing to regularize near small x , or using an enum layout that causes AH or source-term code to read the wrong variable.

8 Unresolved Before Coding

Before implementation, the project still needs a staged CCZ4 modified-cartoon RHS derivation. The future sub-plan is to derive reduced metric and inverse-metric bookkeeping including h_{ww} and γ_{ww} , derive reduced Christoffels including all $1/x$ terms and regular limits, derive hidden Ricci tensor/scalar contributions, derive conformal-metric RHS terms involving h_{ww} , derive extrinsic-curvature RHS terms involving A_{ww} and trace terms, derive Gamma/connection and constraint-propagation terms, derive gauge-equation terms and verify whether hidden-direction contributions enter, and validate each block against symbolic calculations and exact test data before coding the next block.

The project also still needs symbolic verification, a small- x regularization decision, the exact `hww/Aww` enum layout, a reliable `GR_SPACEDIM = 4` build path, unit-test definitions, direct comparison to spherical-coordinate formulas, and convergence acceptance criteria.